

Database Management System

BCA — Semester IV — 2025 — Answer Sheet

Subject Code: CACS255

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. What is flat file system? Explain the advantages of using DBMS over flat file system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. How logical data independence is different from physical data independence? Explain three level database architecture.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. How primary key is different from super key? Explain composite key and partial key.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What are selection and projection operators in relational algebra? Given a table with schema Student(Rollno(PK), Fname, Lname, Address, Level, Did(FK)), Department(Did(PK), Dname, HOD) Write an RA query to find the name of students who are in 'Bachelor' level living in 'Birgunj' and studies under the department with department head 'Ramesh Sharma'.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. How parameters can be passed to a stored procedure? Explain with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. How timestamp based algorithm uses timestamp to serialize the execution of concurrent transactions? Explain in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What are different types of indexing used in database? How index can be created using SQL? Write with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. What are the advantages of normalization? Explain 3NF and BCNF with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. How ER model can be mapped to the relational data model? Explain with a proper example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. a) What is inner join? Perform inner join to find the customer names who are going on flight to 'South Wales':
Customer (Cid, Name, Address, Age, Email) Booking(Cid(FK), FlightID(FK), BookedDate) Flight(FlightID, AirplaneCode, DepartureTime, Destination, Duration)
b) What are unary and ternary relationship? Write with examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.